

RAID ARCHITECTURE

DBMS REVISION & QUICK REFERENCE NOTES

GATE / INTERVIEW READY

1. FUNDAMENTAL CONCEPTS

RAID stands for **Redundant Array of Independent Disks** (originally *Redundant Array of Inexpensive Disks*). It combines multiple physical disk drives into a single logical unit to achieve performance, fault tolerance, or both.

⚡ Why Do We Need RAID?

- **Performance:** Faster read/write throughput via parallel I/O operation across disks.
- **Availability:** Keeps services operational even when physical storage media fail.
- **Data Protection:** Prevents catastrophic data loss in high-scale systems (e.g., web servers).

🧠 Terminology Breakdown

Term	Meaning
Redundant	Extra copy or duplicated data
Array	Collection of multiple disks
Independent	Disks operate separately

2. CORE RAID BUILDING BLOCKS

♦ Striping

Divides data into blocks/pieces and writes them across multiple disks in parallel.

Goal: High Performance

♦ Mirroring

Maintains an exact identical copy of data on a separate secondary disk.

Goal: High Availability

♦ Parity

Calculates redundant information (e.g., XOR) to reconstruct data after disk failure.

Goal: Data Recovery

3. STANDARD RAID LEVELS BREAKDOWN

● RAID 0 – Striping

Speed

Data is split across disks without redundant parity or backup.

Data: Block A, B, C, D

Disk 1	Disk 2
A1	A2
A3	A4

Advantage Concurrent I/O → Max Performance

Drawback 0 Fault Tolerance. 1 failure loses data.

🧠 RAID 0 = Striping = Pure Speed

● RAID 1 – Mirroring

Security

Duplicate copies of identical data are kept on separate disks.

Data Copying

Disk 1	Disk 2
A	A (Copy)
B	B (Copy)

Advantage High protection. 1 disk failure safe.

Drawback 50% capacity overhead (expensive).

🧠 RAID 1 = Mirror = Security

⚡ NESTED & PARITY-BASED RAID LEVELS

● RAID 1+0 (RAID 10) – Nested Striping & Mirroring

Speed + Safety

Combines RAID 1 (Mirroring) and RAID 0 (Striping). Data is striped across mirrored disk pairs.



Use Cases: High-traffic transactional systems, database storage, email and web servers.

Key Benefit: High performance alongside fast, simple recovery from disk failures.

● RAID 2 – Bit-Level

Uses bit-level striping with Hamming code for error correction.

Status: Obsolete and unused in modern commercial systems.

● RAID 3 & 4 – Dedicated Parity

Divides data across disks and stores parity on **one single dedicated disk**.

Bottleneck Every write operation forces an update to the dedicated parity disk, creating a severe bottleneck.

💡 How Parity Reconstruction Works (Simplified Example)

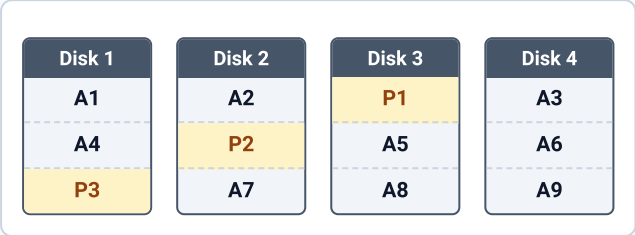
Suppose Data = [1, 2, 3]. Parity = 1 + 2 + 3 = 6.

If Disk 1 fails (Data '1' lost): Missing Value = Parity (6) – Data 2 (2) – Data 3 (3) = 1 (Recovered!).

● RAID 5 – Distributed Parity

1-Disk Fault Tolerant

Distributes parity blocks uniformly across *all* drives, solving the single-parity bottleneck.



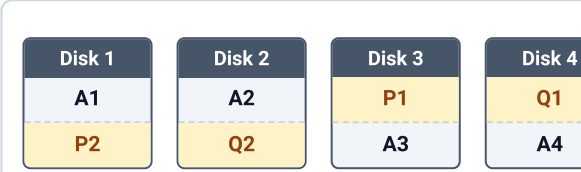
Key Benefit: Solves write-bottlenecks. Tolerates 1 disk failure.

🧠 RAID 5 = Distributed Parity

● RAID 6 – Double Parity

2-Disk Fault Tolerant

Calculates **two independent parity blocks** (P+Q) per stripe distributed across disks.



Key Benefit: Protects against 2 simultaneous disk failures using independent math equations.

🧠 RAID 6 = Double Parity

📋 4. COMPREHENSIVE COMPARISON MATRIX

RAID Level	Main Technique	Primary Purpose	Fault Tolerance	Min Disks
RAID 0	Striping	⚡ Maximum Performance	None (0 disks)	2
RAID 1	Mirroring	🛡️ Data Protection & Reliability	1 Disk Failure	2
RAID 1+0	Striping + Mirroring	⚡🛡️ Speed + High Availability	1 Disk per Mirror Pair	4
RAID 2	Bit-level Striping	Historical Interest	Hamming Code	Obsolete

 5. CRITICAL EXAM TRAPS & COMMON MISCONCEPTIONS

TRAP 1: "RAID 0 provides data redundancy"

FALSE. RAID 0 focuses strictly on performance through striping. It offers zero fault tolerance.

TRAP 2: "RAID 10 requires ten disks"

FALSE. RAID 10 means **RAID 1 + RAID 0** (Nested RAID). It requires a minimum of 4 disks.

TRAP 3: "RAID 1 uses striping"

FALSE. RAID 1 relies entirely on **Mirroring** (duplication).

TRAP 4: "RAID 5 has a dedicated parity disk"

FALSE. Dedicated parity is a feature of RAID 3 and 4. RAID 5 **distributes** parity across all disks.

 6. HIGH-YIELD INTERVIEW Q&A

Q1. What is the fundamental difference between RAID 3 and RAID 5?

RAID 3 uses a single dedicated parity disk which introduces a severe write bottleneck. RAID 5 distributes parity evenly across all drives to balance write load.

Q2. Why is RAID 6 chosen over RAID 5 in large enterprise environments?

RAID 6 maintains two separate parity calculations, allowing the system to survive two concurrent disk failures—crucial during lengthy disk rebuild operations.

Q3. What is Parity and how does it aid data recovery?

Parity is redundant mathematical data (typically calculated via XOR) stored alongside normal data. If one disk fails, the missing values are recalculated using the remaining data and parity.

Q4. What is nested RAID?

Nested RAID combines two basic RAID levels to gain benefits of both. For instance, RAID 1+0 combines RAID 1 (Mirroring) for safety with RAID 0 (Striping) for speed.

 7. 1-MINUTE RAPID RECALL CHEAT SHEET

Hierarchy Tree

RAID

- 0 → Striping (Speed)
- 1 → Mirroring (Safety)
- 10 → Striping + Mirroring
- 2 → Bit-Level (Obsolete)
- 3/4 → Dedicated Parity
- 5 → Distributed Parity
- 6 → Double Parity

 Golden Memory Rules

- 0 = Speed (Striping)
- 1 = Mirroring (Protection)
- 10 = Speed + Security
- 5 = Distributed Parity (1 Disk Fail)
- 6 = Double Parity (2 Disk Fail)

